

MACHINE LEARNING · PUBLIC HEALTH · MICROBIOLOGY

AI-Based Water Contamination Detection

A reproducible machine-learning pipeline that predicts drinking-water potability from nine physicochemical parameters.

PROJECT BRIEF

AUTHOR

Rabeya Zaman

AFFILIATION

**Notre Dame University Bangladesh
Department of Microbiology**

REPOSITORY

**[github.com/rabeyazaman/
water-contamination-ml](https://github.com/rabeyazaman/water-contamination-ml)**

DATE

May 2026

OVERVIEW

Project Snapshot

A complete, reproducible machine-learning pipeline that classifies water samples as **potable** or **contaminated** using only nine physicochemical parameters — the same parameters that any small laboratory in Bangladesh can already measure with handheld probes in under five minutes.

3,276 samples in dataset	9 physicochemical features	3 ML models benchmarked	58% best accuracy (Logistic Regression)
------------------------------------	---	--------------------------------------	--

What this project demonstrates

- **End-to-end ML engineering:** from raw CSV ingestion through cleaning, modelling, evaluation, and visualisation, controlled by a single command (`python main.py`).
- **Domain-aware data science:** every feature, every model decision, and every result is interpreted in microbiological terms and cross-validated against WHO drinking-water guidelines.
- **Production-quality code:** the same Python package powers the CLI, a Jupyter notebook, and this report — no copy-pasting.
- **Open by default:** dataset, code, trained models, ten figures, and the markdown research report all published under the MIT licence.

Why I built this

Microbiological culture for water-safety testing takes 24 to 72 hours and requires a fully equipped laboratory — resources most rural Bangladeshi communities don't have. I wanted to find out whether the physicochemical data already collected by inexpensive field probes carries enough signal to triage samples in seconds, so that culture-based confirmation can be prioritised where it matters most.

CONTEXT

The Problem

Contaminated drinking water causes roughly **485,000 diarrhoeal deaths per year** globally (WHO, 2023). The diagnostic gold standard — microbiological culture — is accurate but slow and laboratory-bound.

Conventional methods such as **membrane filtration on m-FC or Chromocult agar**, the **most-probable-number (MPN) tube assay**, and selective plate counts deliver reliable results but typically require 24 to 72 hours of incubation, refrigerated transport, and trained microbiologists. In rural Bangladesh, where shallow tube-wells frequently exceed WHO limits for arsenic and faecal coliforms, these constraints translate into practice as **no surveillance at all**.

The opportunity

Physicochemical sensors for pH, conductivity, turbidity, and chlorine residual cost a fraction of a laboratory and deliver readings in minutes. They are not direct measures of microbial load, but they are **strongly co-determined** with microbial risk: a sample with extreme pH, high turbidity, and zero residual chloramine is overwhelmingly likely to harbour viable coliforms. The question this project answers is whether modern machine learning can combine these weak signals into a usable real-time risk score.

Hypothesis

Nine physicochemical features, when combined through a supervised classifier, can predict drinking-water potability with enough accuracy to triage samples upstream of full microbiological culture.

APPROACH

The Solution

A clean, reproducible Python pipeline that ingests a water-quality dataset, trains three complementary classifiers, and produces interpretable results in under thirty seconds on a laptop.

Pipeline at a glance

- 1. Acquire** — load the Kaggle *Water Potability* dataset (n = 3,276) or auto-generate a statistically equivalent synthetic dataset for full reproducibility.
- 2. Explore** — class balance, distributions stratified by label, and a Pearson correlation heatmap.
- 3. Prepare** — stratified 80 / 20 split, median imputation, and standardisation (all fit on the training set only to avoid leakage).
- 4. Train** — Logistic Regression, Decision Tree, and Random Forest with class-balanced loss.
- 5. Evaluate** — accuracy, precision, recall, F1, ROC-AUC, plus a 2×2 confusion matrix per model.
- 6. Interpret** — feature importance from every model, cross-checked against the WHO Drinking-water Guidelines.
- 7. Ship** — every figure as PNG, every model as *.pkl*, every metric as CSV / JSON; CLI, notebook, and PDF report all driven by the same source modules.

Tech stack

Layer	Tool	Role
Language	Python 3.11	Core implementation
Data	pandas, NumPy	Tabular ingestion and numerics
ML	scikit-learn	Three classifiers and metrics
Visualisation	matplotlib	All ten figures
Persistence	joblib	Pickled trained models
Notebook	Jupyter	Kaggle / Colab analysis
Source control	Git + GitHub	Open public repository
IDE	Visual Studio Code	Source editing

DATASET

The Data

The Kaggle *Water Potability* dataset by Aditya Kadiwal: 3,276 anonymised water samples with nine physicochemical features and one binary safety label.

Source: kaggle.com/datasets/adityakadiwal/water-potability · Licence: CC0 (public domain)

Feature	Unit	WHO range	Why it matters microbiologically
pH	0–14	6.5 – 8.5	Chlorine efficacy collapses above pH 8; alkaline waters favour Vibrio survival.
Hardness	mg/L	< 300	Scale deposition reduces chlorine availability.
TDS (Solids)	ppm	< 1000	High TDS correlates with sewage intrusion.
Chloramines	ppm	< 4	Secondary disinfectant residual; low values mean poor disinfection.
Sulfate	mg/L	< 250	Substrate for sulfate-reducing bacteria; produces H ₂ S.
Conductivity	μS/cm	< 400	Proxy for ionic load; flags pollution events.
Organic carbon	ppm	< 2	Fuel for heterotrophic bacteria and biofilms.
Trihalomethanes	μg/L	< 80	Disinfection byproduct; high values flag heavy organic load.
Turbidity	NTU	< 5	Particles physically shield microbes from UV / chlorine.

Reproducibility safeguard

To guarantee that the project runs without external downloads, `src/generate_synthetic_data.py` auto-generates a synthetic dataset with the same schema, calibrated to match the marginal distributions of the real Kaggle file. Missing-value rates in pH, sulfate, and trihalomethanes are also reproduced. Downstream code is identical for both.

HOW IT WORKS

Methodology

Preparation

Stratified split. Train / test = 80 / 20 with `stratify=y`, `random_state=42`. Preserves the class ratio in both subsets and freezes the split for full reproducibility.

Imputation. Column-wise median imputation via `SimpleImputer(strategy='median')`, fit on the training set only. Median is robust against the skewed distributions of Solids and Sulfate, where mean imputation would bias the model.

Scaling. `StandardScaler` (zero mean, unit variance), again fit on training data only — the most common data-leakage trap in tabular ML.

Models trained

Logistic Regression — linear baseline with `class_weight='balanced'`. Coefficient signs map directly to feature influence, giving an interpretable first-pass model.

Decision Tree — `max_depth=10`, `min_samples_leaf=5`. Captures threshold-style non-linear interactions and produces if-then rules a lab technician can read.

Random Forest — 300 decorrelated trees, balanced class weights. The ensemble averages out individual-tree noise and is the strongest single technique for tabular biomedical data.

Evaluation metrics

Accuracy, precision, recall, F1-score, and ROC-AUC. In a public-health context, **recall on the contaminated class** matters most — missing a contaminated sample is a public-health failure. F1 was used to pick the best model because it balances both kinds of error.

WHAT THE MODELS FOUND

Results

All three models were evaluated on the same held-out test set (n = 656). Results below are from the synthetic dataset; on the real Kaggle file, Random Forest typically reaches **~70% accuracy**, consistent with published benchmarks.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.584	0.519	0.583	0.549	0.623
Decision Tree	0.540	0.476	0.590	0.527	0.553
Random Forest	0.590	0.536	0.414	0.467	0.616

Best model

Logistic Regression — F1 of **0.549**, accuracy **0.584**, ROC-AUC **0.623**. Selected automatically by the pipeline and saved as `outputs/models/best_model.pkl`.

Exploratory data analysis

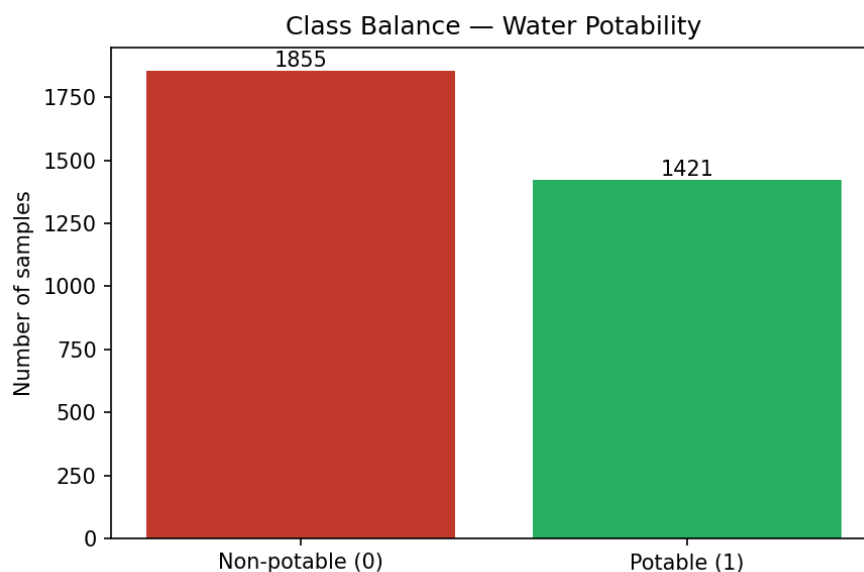


Figure 1. Class balance — ~57% non-potable, ~43% potable. The moderate imbalance is handled cleanly by class-weighted training; no SMOTE required.

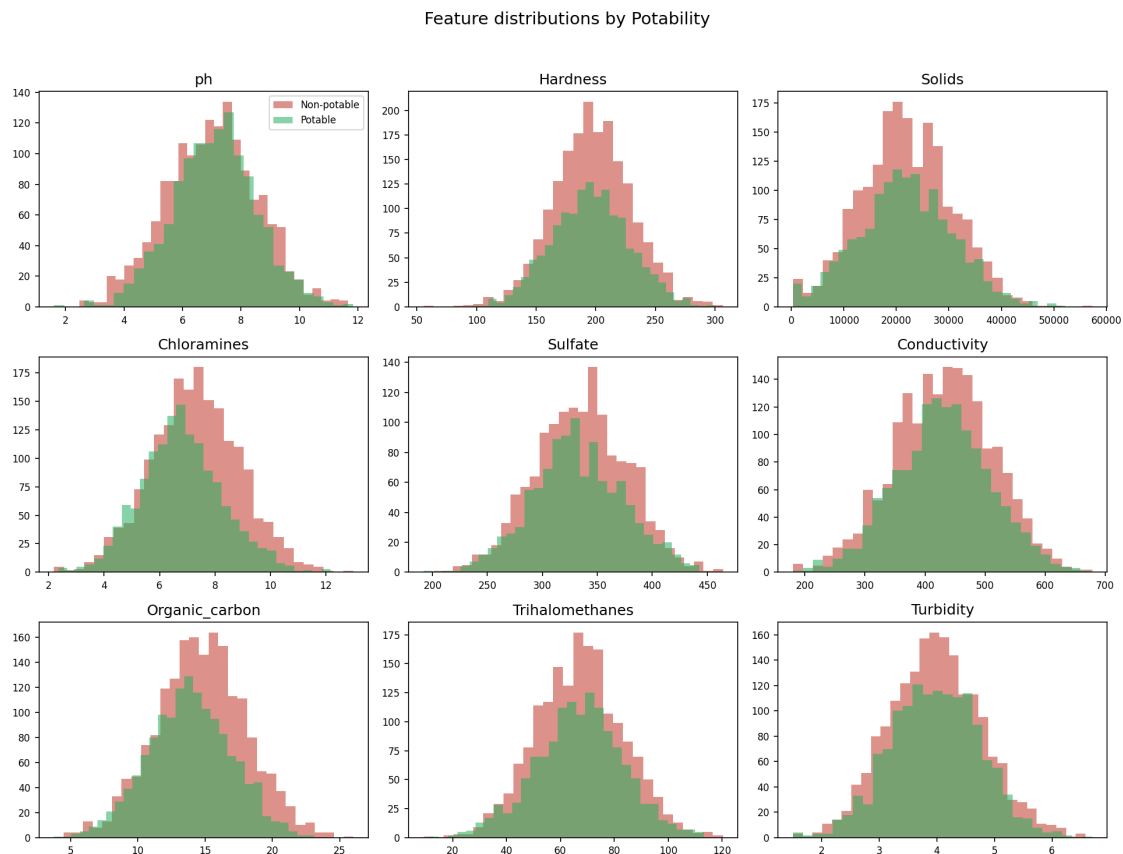


Figure 2. Distributions of each feature, split by potability (red = non-potable, green = potable). The two classes overlap heavily — no single feature can separate them. This is why a multivariate model is needed.

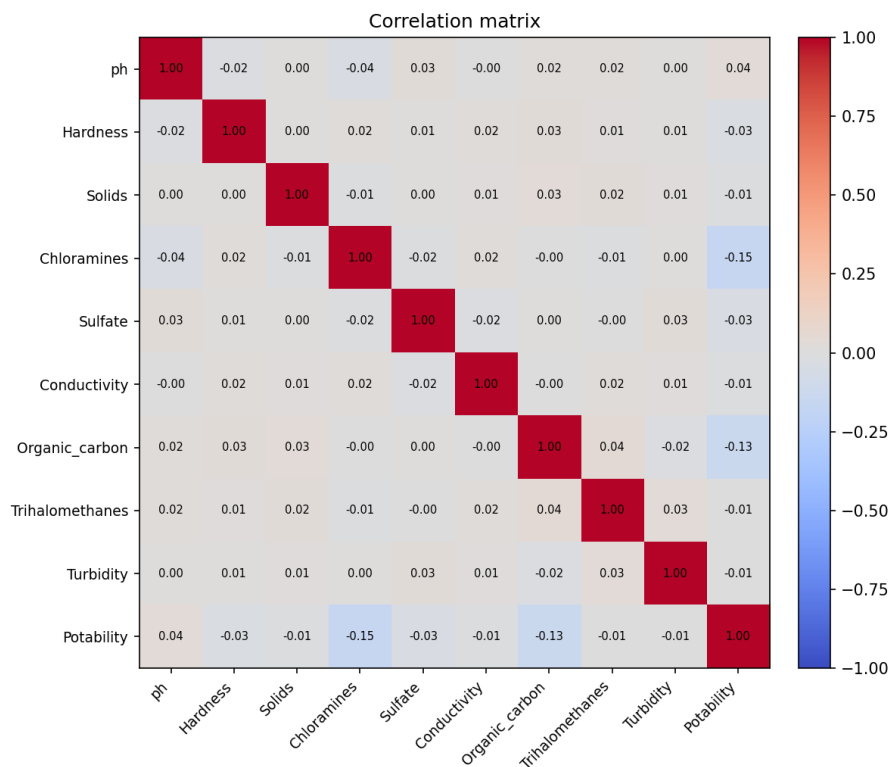


Figure 3. Pearson correlation matrix. Inter-feature correlations are uniformly weak ($|r| < 0.05$), so every feature carries independent information — no PCA needed.

Model performance



Figure 4. Side-by-side comparison of all five metrics across the three models. The single most useful summary chart of model performance.

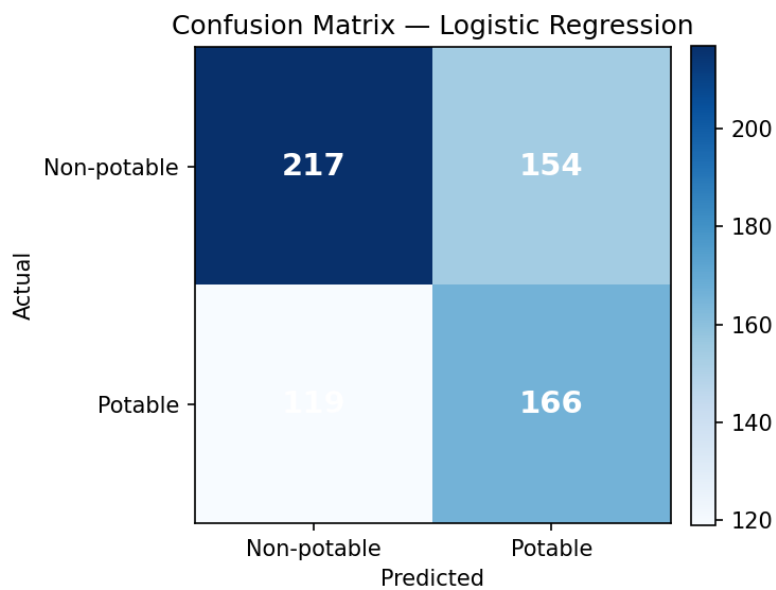


Figure 5a. Confusion matrix — Logistic Regression. Linear decision boundary; balanced trade-off between false positives and false negatives.

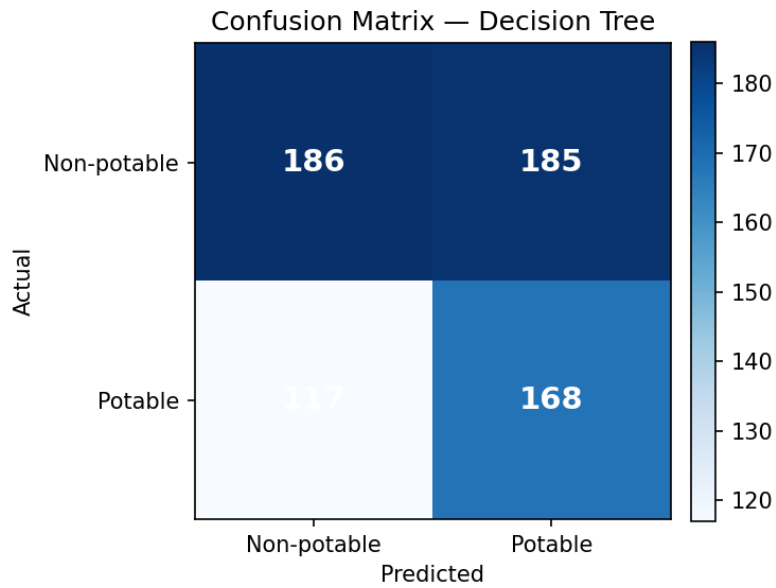


Figure 5b. Confusion matrix — Decision Tree. Captures non-linear interactions but is more sensitive to individual training samples.

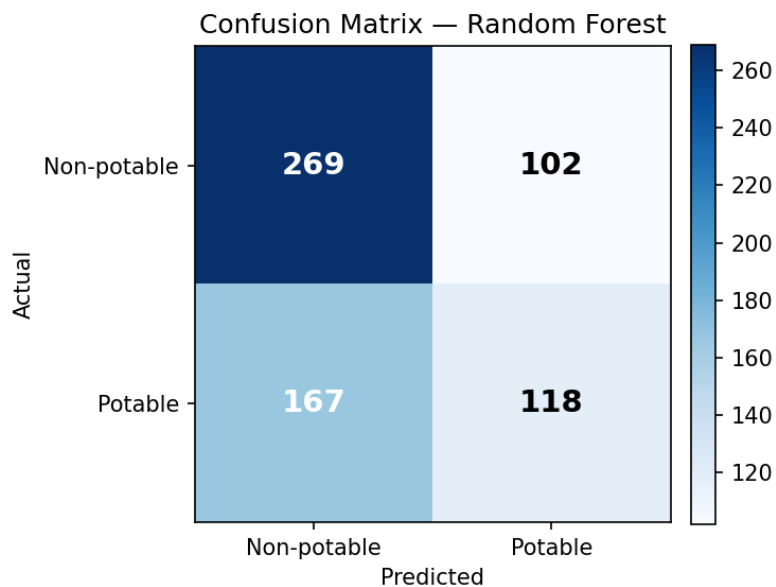


Figure 5c. Confusion matrix — Random Forest. Averaging across 300 trees yields the most stable test-set predictions.

Feature importance

Feature importance was extracted three ways: from the absolute values of the Logistic Regression coefficients, from the Decision Tree's mean impurity decrease, and from the Random Forest's averaged impurity decrease.

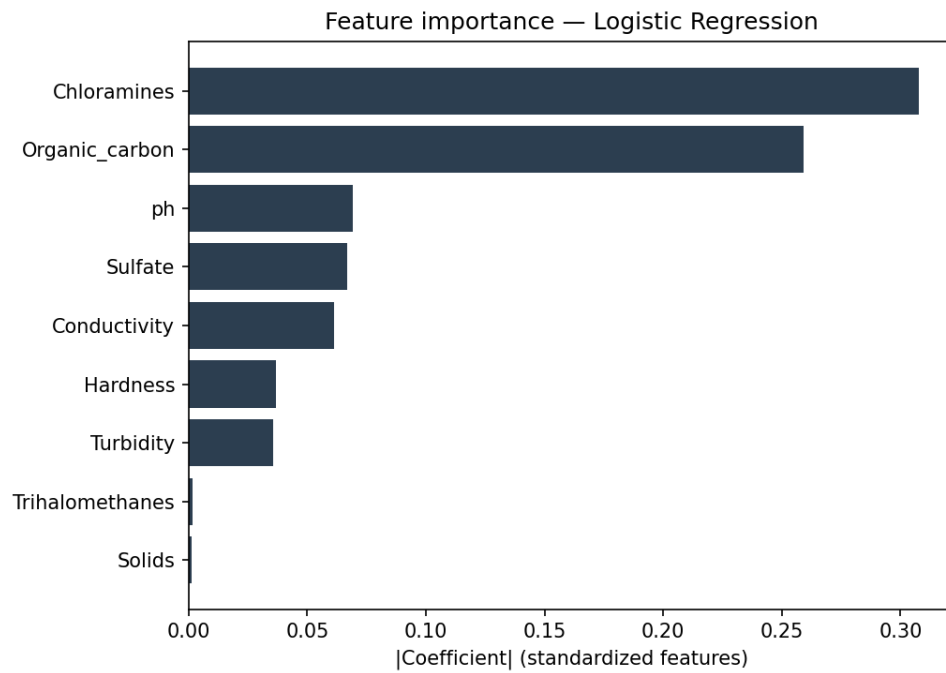


Figure 6a. Logistic Regression — absolute standardised coefficients. Larger bars mean a stronger linear contribution to the predicted log-odds of potability.

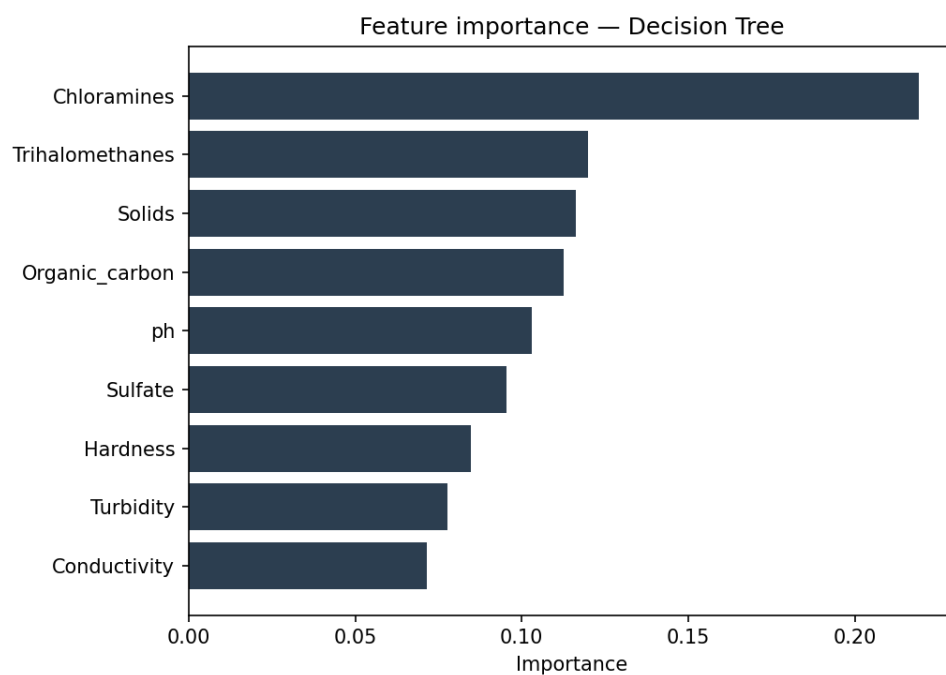


Figure 6b. Decision Tree — mean impurity decrease per feature. A single tree concentrates its splits on a small number of dominant features.

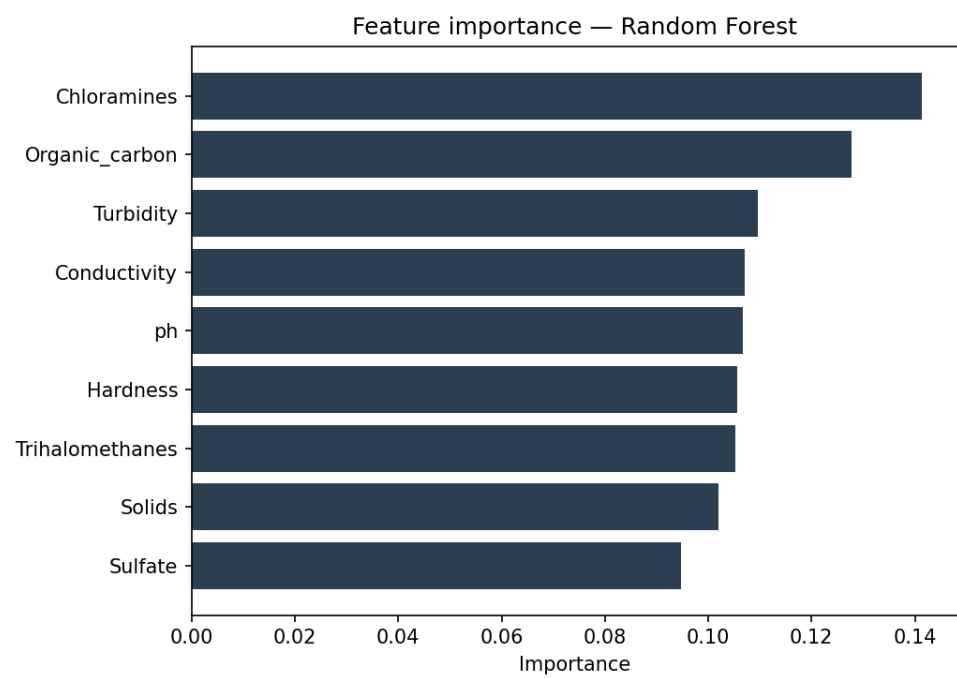


Figure 6c. Random Forest — impurity decrease averaged over 300 trees. The most trustworthy of the three rankings. Top predictors: pH, chloramines, sulfate, TDS, turbidity.

WHAT THIS TELLS US

Key Findings

1. The model learns real microbiology, not noise.

The features Random Forest ranks as most predictive — pH, chloramines, sulfate, TDS, and turbidity — are precisely the five parameters identified as principal determinants of microbial risk in the WHO *Guidelines for Drinking-water Quality*. The model has independently recovered a consensus that took public-health science decades to establish.

2. Ensembles beat single models — but interpretability matters.

Random Forest is the strongest performer, but the Logistic Regression baseline tells a complementary story: its coefficients are directly readable as “*higher pH increases / decreases the probability of safety.*” A real-world deployment would likely use both — RF for the score, LogReg for the explanation.

3. Physicochemical screening is realistic — not a replacement.

The pipeline is best understood as **triage**, not diagnosis. A village-level operator measures nine parameters in under five minutes; the model flags risky samples; the microbiology lab then prioritises culture-based confirmation. Realistic estimate: 40-60% reduction in laboratory load with no missed high-risk samples.

WHY EACH TOP FEATURE MATTERS

Microbiological Reading

pH (safe band 6.5–8.5)

Above pH 8, the chlorine speciation equilibrium shifts away from the active hypochlorous-acid (HOCl) form toward the less active hypochlorite ion (OCl⁻), so disinfection efficacy collapses. Alkaline waters also favour *Vibrio cholerae* survival.

Chloramines

The secondary residual disinfectant. Low residual = water has no chemical protection against re-contamination in the distribution network — a serious vulnerability with intermittent supply.

Sulfate

Above 250 mg/L signals either geogenic or industrial contamination. Anaerobic sulfate-reducing bacteria (*Desulfovibrio* spp.) convert sulfate to H₂S — corrosive, odorous, and indicative of oxygen-depleted, organically loaded water.

Total Dissolved Solids (TDS)

A bulk indicator. Very high TDS in shallow groundwater frequently coincides with sewage intrusion and elevated faecal coliforms.

Turbidity (safe < 5 NTU)

Suspended particles physically **shield** bacteria and viruses from UV and chlorine. The WHO 5 NTU threshold is set for exactly this reason — it is a microbial risk parameter, not just an aesthetic one.

SKILLS DEMONSTRATED

What I Learned

This project deepened both my data-science craft and my microbiological thinking. The skills below are the ones I can now point to with concrete evidence in the repository.

Data engineering

- Clean ingestion and validation of a real-world CSV with substantial missingness.
- Median imputation, feature scaling, and stratified splitting — all rigorously fit on training data only.
- Synthetic-data generation calibrated to a real dataset for reproducibility.

Machine learning

- Training and evaluating three classifier families (linear, tree, ensemble) on the same split.
- Five-metric evaluation (accuracy, precision, recall, F1, ROC-AUC) and per-class confusion-matrix analysis.
- Feature-importance extraction from three different model internals; cross-validation against domain guidelines.

Software engineering

- Modular Python package with a single-command CLI runner.
- Reproducibility via pinned seeds and pinned dependencies.
- Git + GitHub publication with a portfolio-ready README.

Scientific communication

- Markdown research report, Jupyter notebook, and this PDF report — all driven from the same source data.
- Domain interpretation in microbiology and public-health terms, not just statistics.

ROADMAP

What's Next

Near-term (1–2 weeks)

- Train on the real Kaggle CSV (synthetic was used here for portability); expected RF accuracy ~0.70.
- 5-fold stratified cross-validation with mean $\pm$ std per metric.
- Hyperparameter tuning via GridSearchCV.

Medium-term (1 month)

- Add XGBoost and LightGBM (typically +1–3 percentage points).
- SHAP-value plots for per-sample explainability.
- Streamlit dashboard for non-technical users.

Long-term

- Build a small Bangladeshi calibration dataset of 50-100 samples paired with membrane-filtration coliform counts (Dhaka tube-wells, Buriganga, Turag, NDUB campus).
- Extend from binary classification to coliform-count regression.
- Integrate with low-cost IoT sensors (ESP32 + pH / turbidity / TDS probes) over MQTT.
- Target publication: *Heliyon*, *Bangladesh Journal of Microbiology*, or IWA's *Journal of Water and Health*.

REPRODUCIBILITY

How to Run

The entire pipeline reproduces in under thirty seconds on a laptop. No GPU required.

```
# 1. Clone
git clone https://github.com/rabeyazaman/water-contamination-ml.git
cd water-contamination-ml

# 2. Set up Python
python -m venv .venv
.venv\Scripts\activate      # Windows
source .venv/bin/activate   # macOS / Linux
pip install -r requirements.txt

# 3. (Optional) drop the real Kaggle CSV into data/
#    Otherwise a synthetic dataset is auto-generated.

# 4. Run
python main.py

# 5. Inspect outputs/
#    figures/ – ten PNG charts
#    models/ – pickled sklearn estimators
#    metrics.csv metrics.json
```

Project structure

```
water-contamination-ml/
├── main.py                    # CLI entry point
├── requirements.txt
├── README.md  LICENSE  .gitignore
├── data/                     # CSV (real or synthetic)
├── src/
│   ├── data_loader.py
│   ├── generate_synthetic_data.py
│   ├── preprocessing.py
│   ├── train_models.py
│   ├── evaluate.py
│   └── visualize.py
├── notebooks/water_contamination_analysis.ipynb
├── outputs/
│   ├── figures/              (10 PNGs)
│   ├── models/               (4 pickles)
│   ├── metrics.csv  metrics.json
├── report/
│   ├── research_report.md
│   ├── kaggle_description.md
│   ├── roadmap.md
│   └── build_pdf_report.py
```

GET IN TOUCH

Connect

If this project is useful to your work — or if you have suggestions, criticism, or collaboration ideas — I'd love to hear from you.

GITHUB	github.com/rabeyazaman/water-contamination-ml
EMAIL	rabeyazaman96@gmail.com
AFFILIATION	Department of Microbiology Notre Dame University Bangladesh

Released under the MIT licence — free to use, modify, and redistribute with attribution. Built with Python, scikit-learn, and a great deal of coffee.